

Coupon Distributer Mechanism Purpose and Process Description

The function of this system is to transfer a stack of vertical coupons one by one through: a selector using vacuum, a transfer gear – shaft system that sucks coupons from a vacuum plate using fine adjustment to set the vacuum supplied, and one pair of driving arms that transfers the coupons from the vacuum plate to the transferring rollers through a main drive shaft speed of 270 rpm with safety operating speed of 270 rpm. The main drive shaft's speed derives its speed from a main timing belt with respect to gear ratios.

The entire coupon unit of this system can be summarized by the following Free Body Diagrams and their corresponding equations.

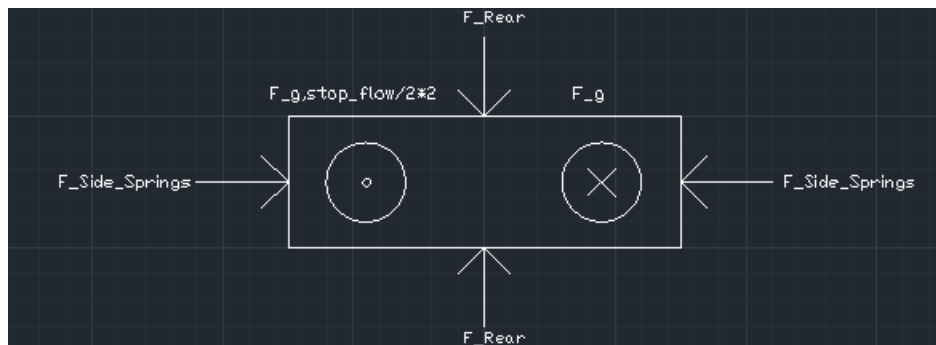


Figure 1: Free Body Diagram of the Coupon Stack Selector Mechanism and Force Balance

When the stack of coupons is loaded into the unit magazine according to the manual a Rear Coupon Guide (12) is built to supply a small force F_{Rear} is exerted between the front and back guide to ensure that the coupons do not fall off. Additionally, the stack is also supported by side support springs (17) by exerting forces $F_{Side_Springs}$ to ensure balance along each coupon width. Finally, to implement the coupon selector mechanism. Since the total weight of the stack is represented as the total force of gravity acting on the last coupon, cylinder pair (9) create a combined counteractive force $(F_{g,Stop_Flow}/2)*2$ when their corresponding springs (10) are tightened. With the position of the cylinders regulated by tightening/loosening the screws (11). Finally, each coupon before being transferred to the rollers is in vacuum.

Note: See AutoCAD file Free Body Diagrams

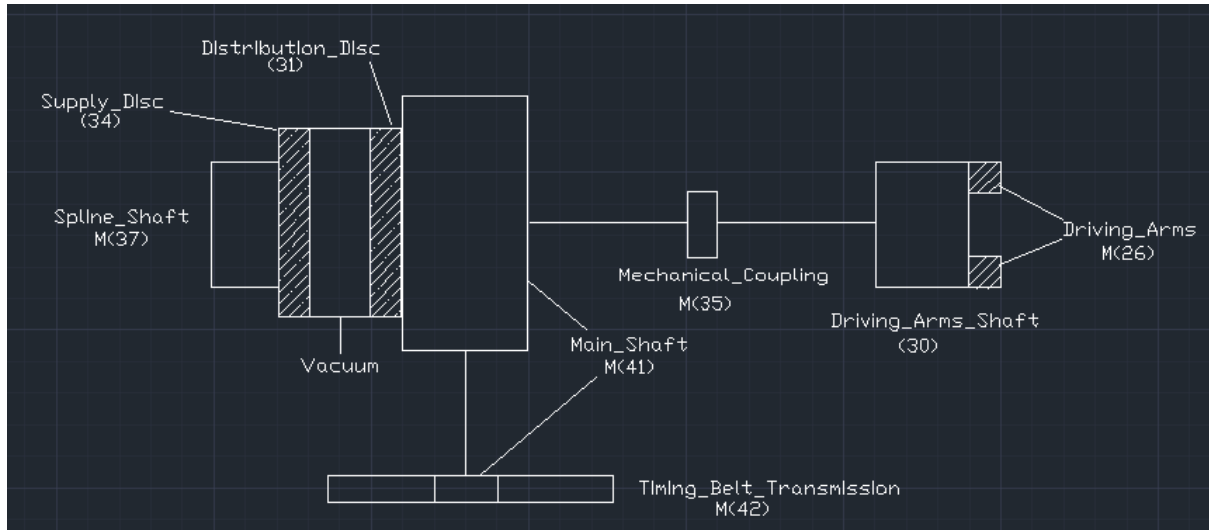


Figure 2: Free Body Diagram of the Moments and Power Transfer Between Shafts Illustrating How An Extracted Coupon is Transferred from Vacuum Plate to the Motion Rollers

Since this system is a subcomponent of a much bigger system. Main Shaft (41) will exert a moment $M(41)$ through a timing belt from a smaller gear to a larger gear as shown above. Implying more torque or power will be transferred for the coupon unit using smaller initial power. The Spline Shaft (37) will be set externally from HMI for either no-coupon operation or coupon operation and as a result there will be a change in moment $M(37)$. The Supply Disc (34) will be loosened while the Distribution Disc (31) is tightened. This controls the amount of vacuum between the vacuum plate (20) and motion rollers linking the vertical slider motion subsystem and this current subsystem with the reciprocating motion rollers. The mechanical power is transferred once again through a Mechanical Coupling creating a change in moment through $M(35)$. Ideally the change in moment should be zero in ideal smooth operation. The Driving Arms Shaft (30) fixes the Driving Arms (26) in place. Each driving arm as a result will have a transferred moment $M(26)/2$. This forces the Driving Arms to work synchronously with the Main Shaft, Vertical Slider Motion Subsystem, Vacuum Creation and linking it to the motion rollers.

The following system of equations summarize how this subsystem works in practice.

$$(1) \sum F = 0 : F_{REAR} - F_{REAR'} + F_{SIDESPRINGS} - F_{SIDESPRINGS'} + F_g - 2 * F_{gSTOPFLOW} / 2 = \lim_{\epsilon \rightarrow 0} \epsilon * F_g$$

$$(2) \sum M = 0 : N_{42} < N_{41}, M_{41} = N_{41} * M_{42} / N_{42}, M_{41} + \Delta * M_{37} + \Delta * M_{35} = 2 * M_{26} / 2$$

The remaining parts of the Coupon Unit Technical Document describe its practical application of the functionality of each screw, why one should loosen or tighten it, what manifolds hold what, when to loosen, when to tighten, under what conditions, etc. But the above represents its main functionality.

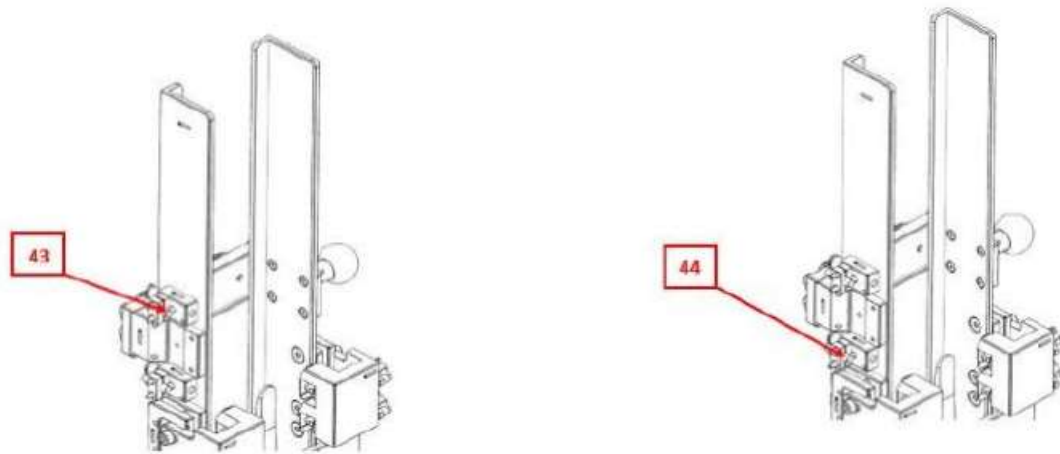


Figure 3: (43) Sensor checks for coupon level in magazine unit under ordinary operation if sensor detects it is below it or if it suddenly doesn't notice it then the coupon unit must be refilled immediately. (44) sensor acts as a safety sensor and acts in emergency situations. If coupon is below that height then operation is no longer safe and must be stopped. All these processes are programmable through some PLC platform.

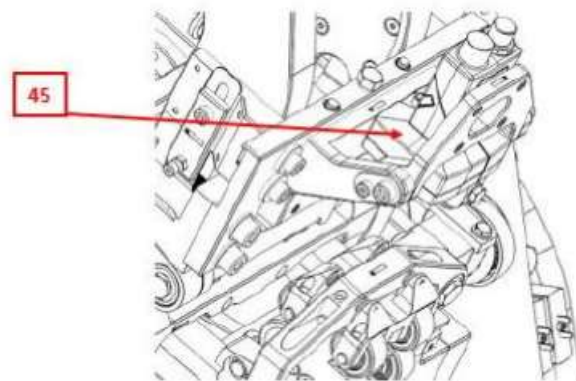


Figure 4: Checks if orientation of coupons is correct.

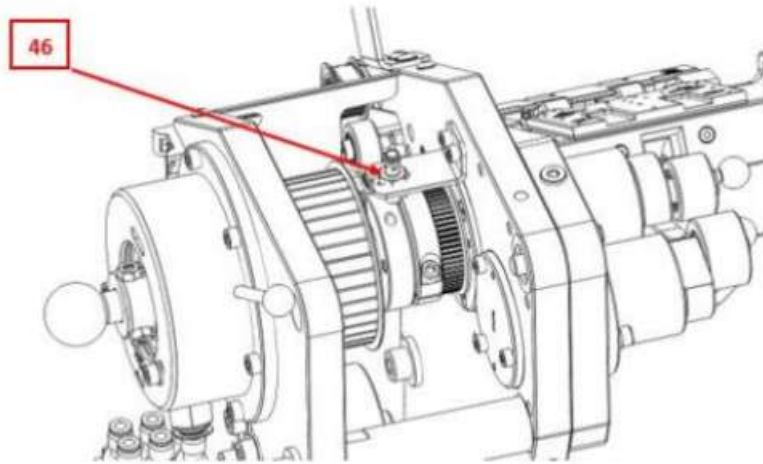


Figure 5: (46) Sensor checks if coupling (35) is adjusted in the correct position to ensure smooth coupling.

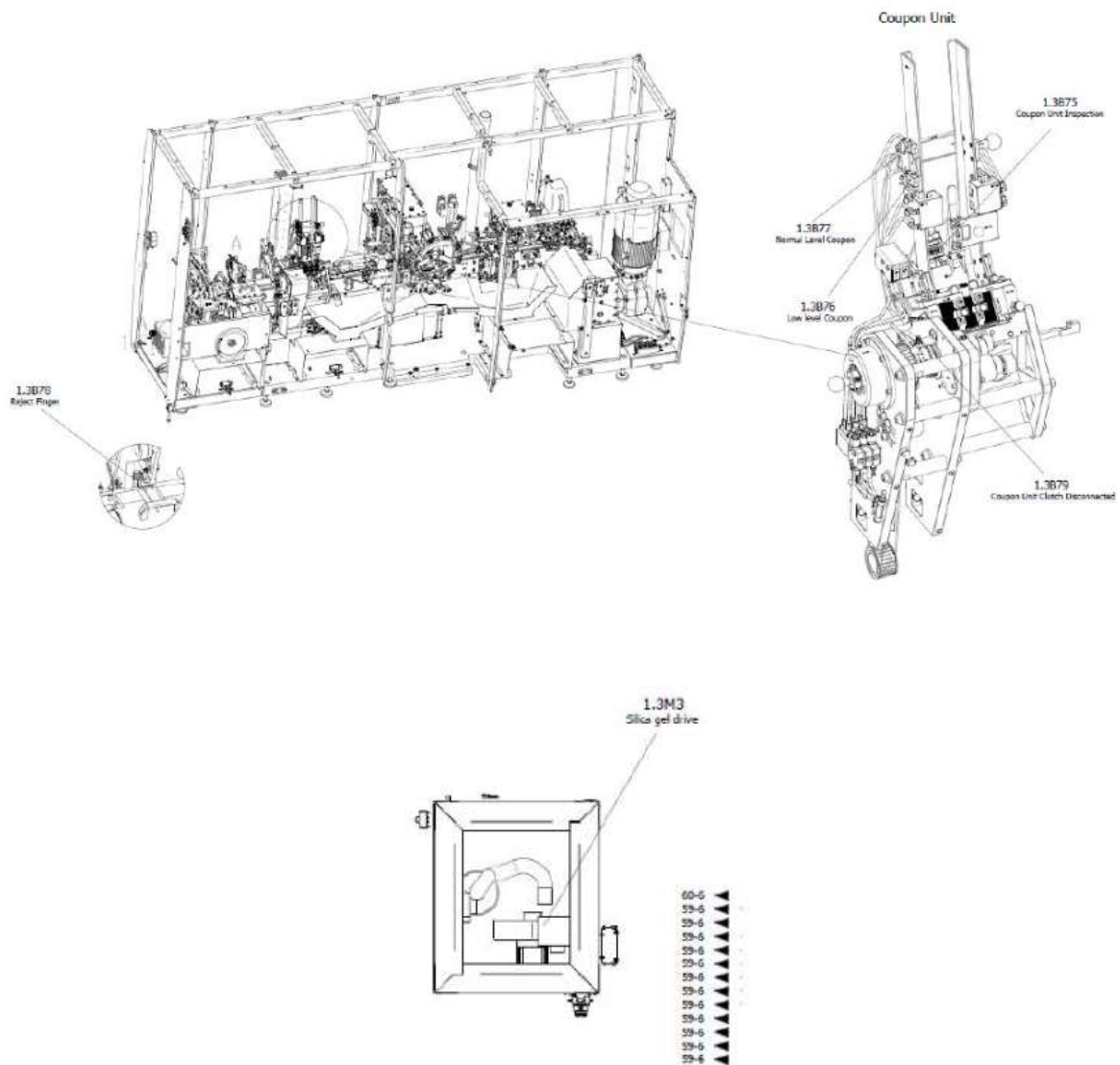


Figure 6: As one can see this subsystem is part of a much larger system. The primary function of the coupon unit in simple terms is to convert y dimensional translational motion to x dimensional translational motion. It then has other components that are activated/deactivated depending on PLC/HMI settings an example is the silica gel driver which could be a possible next step after the coupon unit subsystem component.